① 

$V_0 = 3\,m/s \qquad a = 2\,m/s^2$

$V' = 20\,m/s \qquad x' = ?$

$$X = X_0 + V_0 t + \tfrac{1}{2} a t^2$$

$$x = 3t + \tfrac{1}{2} \cdot 2 t^2 \quad ①$$

$$V = V_0 + at$$

$$\Rightarrow 20 = 3 + 2t$$

$$\Rightarrow 2t = 17$$

$$\Rightarrow t = 8.5\,s \Rightarrow x = 3(8.5) + (8.5)^2$$

$$\Rightarrow x = 97.75\,m \;//$$

Εναλλακτικός τρόπος εύρεσης

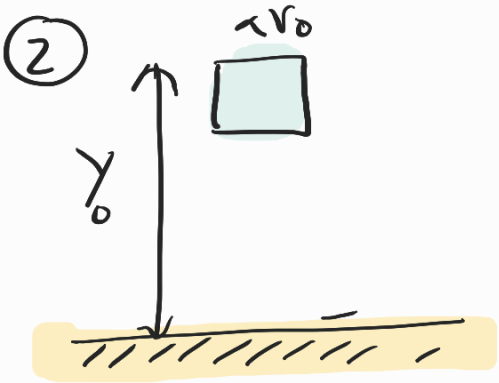
$$X = v_0 t + \tfrac{1}{2} a t^2$$

$$\Rightarrow 2 \cdot a \cdot x = 2 a v_0 t + a^2 t^2 \quad ①$$

$$V^2 = (V_0 + at)^2 \quad = ①$$

$$\Rightarrow V^2 = V_0^2 + 2 V_0 a t + a^2 t^2 \quad ②$$

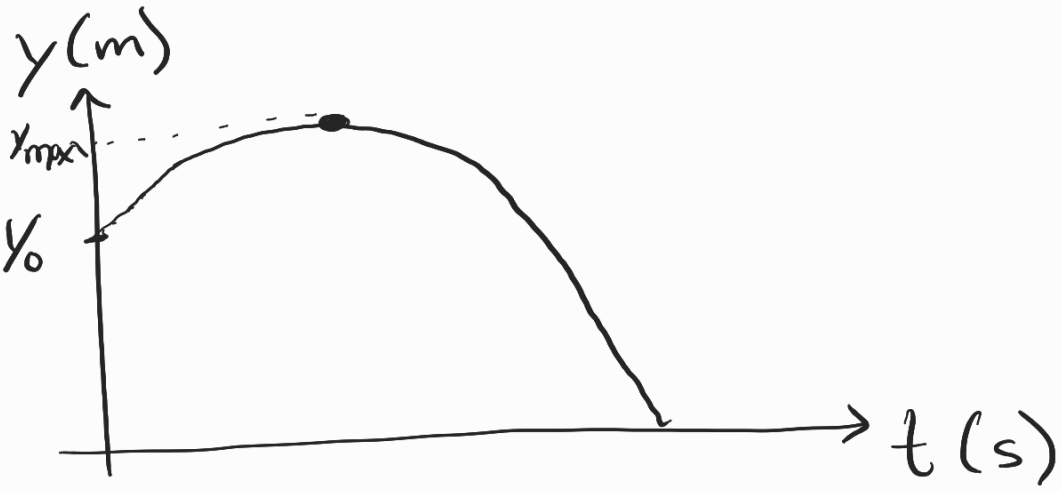
① → ② $\boxed{V^2 = V_0^2 + 2ax}$

② 

$v_0 = 5\,m/s$

$a = -g$

$y_0 = 6m$

(i) 

(ii) $y_{max} = ?$ στο y_{max}, $v = 0$

$v^2 = v_0^2 + 2ay$ ($y = y_{max} - y_0$)

$\Rightarrow v^2 = v_0^2 - 2 \cdot g (y_{max} - y_0)$

↓ δες τη γραφική

$\Rightarrow 0 = 5^2 - 20(y_{max} - 6)$

$\Rightarrow 20(y_{max} - 6) = 25$

$\Rightarrow 4 y_{max} - 24 = 5$

$\Rightarrow y_{max} = 29/4 = 7.25m$

iii) for $y=0$, $t=?$

$$y = y_0 + v_0 t + \frac{1}{2}at^2$$

$$\Rightarrow 0 = 6 + 5t - 5t^2$$

$$\Rightarrow 5t^2 - 5t - 6 = 0$$

formula for quadratic equations

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow t = \frac{5 \pm \sqrt{25 + 4 \cdot 5 \cdot 6}}{10}$$

$$= \frac{5 \pm \sqrt{145}}{10}$$

$$t = \frac{5+12}{10}$$

$$t = 1.7\,s \;//$$

ή

$$t = \frac{5-12}{10}$$

$$= -7/10\,s$$

αρνητικός χρόνος δεν έχει νόημα

iv) $v_f = ?$ $\quad v = v_0 + at$

$t_f = 1.7s \Rightarrow v_f = 5 - 10(1.7)$

$v_f = 5 - 17 = -12 m/s$ //

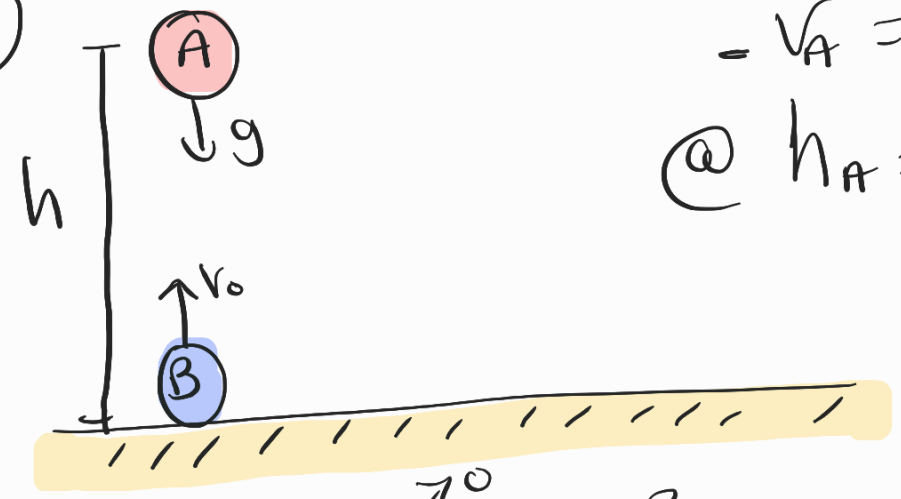
γιατί όχι $v^2 = v_0^2 + 2ax$;

i.e. $v^2 = 0 - 2 \cdot 5 \cdot 11$

$v^2 = -110$!! → λάθος!

γιατί αυτό υπολογίζεται από $y_0 = 0$ ενώ εμείς είχαμε $y_0 = 6m$.

③



$-V_A = 2V_B$ ①

@ $h_A = h_B = xh$

$Y_A = Y_{0A} + \cancel{V_{0A}}^{0} t - \frac{1}{2}gt^2 \Rightarrow Y_A = h - \frac{1}{2}gt^2$ ②

$Y_B = \cancel{Y_{0B}}^{0} + V_{0B}t - \frac{1}{2}gt^2 \Rightarrow Y_B = V_{0B}t - \frac{1}{2}gt^2$ ③

② = ③ $\Rightarrow h - \frac{gt^2}{2} = V_{0B}t - \frac{1}{2}gt^2$

$\Rightarrow h = V_{0B}t$ ④

$V = V_0 + at$

$\Rightarrow V_A = -gt$ và $V_A = -2V_B$

$\Rightarrow V_B = gt/2 = V_{0B} - gt$

$\Rightarrow V_{0B} = \frac{gt}{2} + gt = \frac{3gt}{2}$ ⑤

⑤ → ④

$h = \frac{3gt^2}{2}$

$\Rightarrow \frac{h}{3} = gt^2/2$ ⑥

⑥ → ②

$\Rightarrow Y_A = h - \frac{h}{3}$

$= 2h/3$ //

④  $\rightarrow v_A = 6\ km/h$ Δx_{AB} at $t = 10m$

 $\rightarrow v_B = 4\ km/h$

$t = 10min = 10 \cdot 60 = 600s$

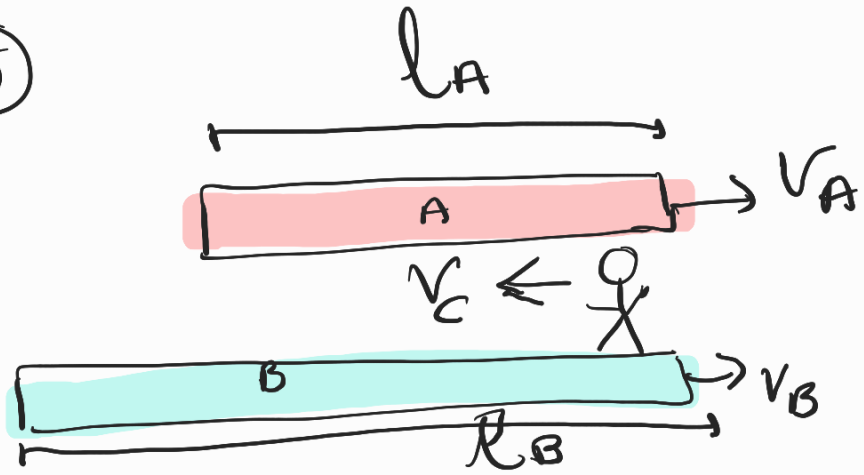
$$v_A = \frac{6km}{h} = \frac{6 \times 10^3 m}{60 \cdot 60} = \frac{60}{6 \cdot 6} = \frac{10}{6} = 1.7\ m/s$$

$$v_B = \frac{4km}{h} = \frac{4 \times 10^3 m}{3600} = \frac{40}{36} = 1.1\ m/s$$

$$v_{AB} = 0.6\ m/s$$

$$\Leftrightarrow \Delta x_{AB} = 0.6 \cdot 600 = 360\ m //$$

⑤



$v_A = 108\ km/h$

$v_B = 72\ km/h$

$v_c = -9\ km/h$

↓

Ο Ε ≡ ανθρώπινος

$l_A = 125m \quad l_B = 180m$

$v_c' = 72 - 9 = 63\ km/h$ → ταχύτητα του Ο ως προς το έδαφος

→ ταχύτητα τρένου ως προς Ο

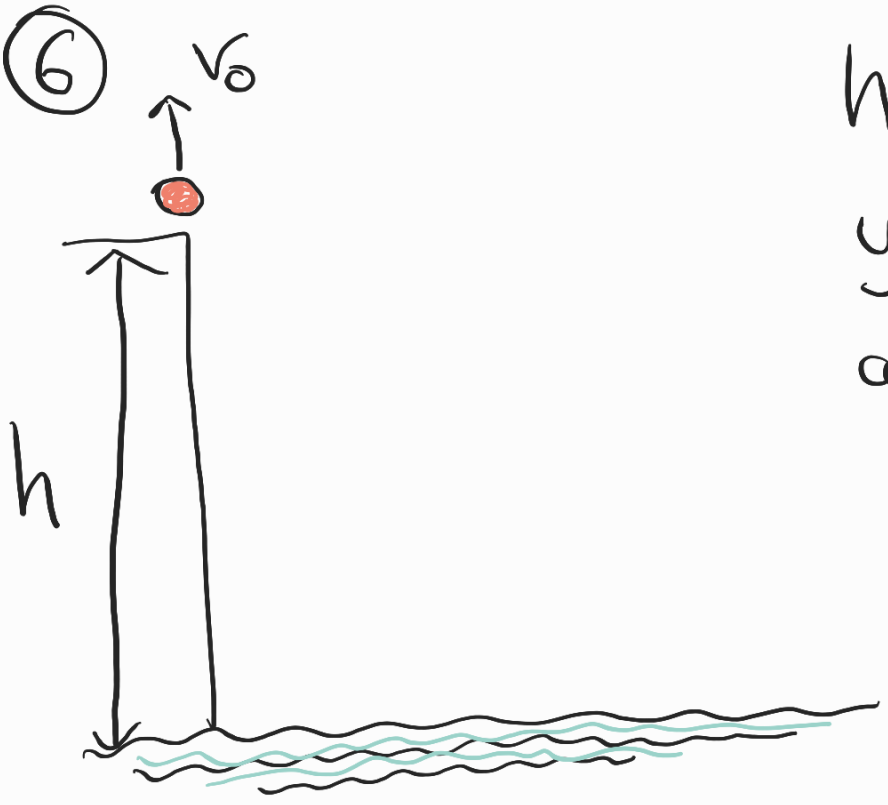
$v_{CA} = 108 - 63 = 45\ km/h = \frac{45 \times 10^3}{3600} = 12.5\ m/s$

Για να περάσει το Α το Β πρέπει να περάσει τουλάχιστον αυτό.

$\Rightarrow \Delta x = 125m$

$\Rightarrow v = \frac{\Delta x}{\Delta t} = \Delta t = \frac{125}{12.5} = 10\ s$ //

⑥



$h = 20m$

$y = h$ @ $t = 3s$

$a = -g \Rightarrow$ EOEK

(i) $v_0 = ?$

$$y = y_0 + v_0 t + \frac{1}{2} a t^2$$

$t = 3s \Rightarrow y = h$

$$\Rightarrow \cancel{h} = \cancel{h} + v_0 \cdot 3 - \tfrac{1}{2} g (3)^2$$

$$\Rightarrow 3v_0 = 4.5g$$

$$\Rightarrow v_0 = 45/3 = 15 m/s //$$

(ii) $v_f = ?$

Χτυπά θάλασσα $\Rightarrow y = 0$

$$y = y_0 + v_0 t + \tfrac{1}{2} a t^2$$

$$\Rightarrow 0 = 20 + (15)t - \tfrac{1}{2} g t^2$$

$$\Rightarrow 0 = 20 + 15t - 5t^2$$

$$\Rightarrow t^2 - 3t - 4 = 0$$

$t \quad -4 \Rightarrow (t-4)(t+1) = 0$

$t \quad +1 \Rightarrow t = 4\,s$ ή $t = -1\,s$

↳ χρόνος που χτυπά στη θάλασσα

$$v = v_0 + at$$

$$\Rightarrow v_f = 15 - 10 \cdot 4$$

$$= 15 - 40$$

$$v_f = -25\,m/s$$

(iii) $t = ?$ για $y = 10 + h$

$$(10+h) = h + v_0 t - \tfrac{1}{2} g t^2$$

$$\Rightarrow 30 = 20 + 15t - 5t^2$$

$$\Rightarrow \cancel{5}t^2 - \cancel{15}^{3}t + \cancel{10}^{2} = 0$$

$$t^2 - 3t + 2 = 0$$

$$t \quad -2$$

$$t \quad -1$$

$$(t-2)(t-1) = 0$$

$$\Rightarrow \underline{t = 2\ s} \text{ και } \underline{t = 1\ s}$$